ChatGPT

Executive Summary

Sri Lanka's tea sector faces mounting **climate threats**: rising temperatures, erratic rainfall and extreme events threaten yields and quality【6†L110-L118】【70†L92-L100】. Recent surveys confirm that climate change is already reducing Sri Lankan tea output (a projected **–12% yield by mid-century** under high-emissions scenarios【70†L92-L100】). At the same time, automation and AI offer new opportunities for **precision tea farming**, but most existing research focuses on **prediction** (e.g. yield, quality, stress) rather than actionable **decision support**. For example, recent studies used remote sensing and ML to estimate seasonal yield【18†L107-L118】 or quality (polyphenols)【46†L81-L90】, or to screen drought tolerance【22†L409-L417】, but none provide an integrated system that translates these predictions into management actions.

We propose a novel **AI-Powered Climate-Adaptive Tea Plantation Decision Intelligence** system to fill this gap. Key innovations include new composite indices (a **Climate Stress Index**, **Harvest Opportunity Score**, and **Tea Operational Readiness Index – TOPRI**) with clear formulas and thresholds, plus a Decision Engine that synthesises data (images, weather, soil, yields) to recommend adaptive actions (irrigation, pruning, harvest timing, etc.) with confidence scores. The design integrates four modules: (1) **Climate & Plantation Stress Assessment**, (2) **Climate Scenario Simulator**, (3) **Harvest Readiness & Scheduling**, and (4) **Adaptive Decision Intelligence with Explainability (SHAP, RAG, rule-based)**. Each module's inputs, outputs and algorithms are carefully defined, and new indices (CSI, HOS, TOPRI) are mathematically specified below.

To implement this system we outline (a) a **data plan** using public datasets (NASA POWER, ERA5 climate reanalysis, SoilGrids soil data, Sentinel/Landsat imagery, tea estate records), synthetic data strategies, and an annotation schema; (b) an **evaluation plan** with metrics (MAE, RMSE, F1-score, calibration), baselines (statistical and ML), cross-validation strategies and expert review, including synthetic intervention tests (e.g. simulating delayed harvest); and (c) an **implementation roadmap** (tasks, effort estimates, hardware/sensor needs, deliverables) illustrated by a Gantt chart. Finally, we identify risks (data gaps, label noise, novelty skepticism) and mitigation strategies, and provide a concise justification of novelty for examiners. The design focuses on **Sri Lankan context** (local climate patterns, TRI guidelines) and draws on international best practice (peer-reviewed literature, FAO/TEACROP guidelines).

1. Literature Synthesis and Gaps

We reviewed ~40 recent publications (2020–2026) on tea agronomy, remote sensing, and AI for tea. Table 1 (below) summarises key papers (title, year, venue, methods, outputs and remaining gaps). In brief:

- **Yield & Quality Prediction:** Several studies fuse **remote sensing (UAV, Sentinel)** and ML for *in-season yield/quality* forecasts. For example, Liu *et al.* (2025) used **Sentinel-2**+UAV imagery with six regression models to estimate seasonal fresh tea yield ($R^2 \approx 0.5$)【18†L107-L118】. Paul and Katharina (2025) developed a deep **spatio-temporal model** (RNN+GNN) combining IoT sensors, satellite and weather data to predict parcel-level yield and quality (with uncertainty estimates)【11†L71-L80】【11†L109-L118】. These improve on independent regression, but still produce point estimates only. Time-series models (Nagpal *et al.*, 2026) found SARIMAX(1,0,0)(0,0,1,12) effectively captures yield seasonality (1985–2022 Sri Lankan data)【41†L64-L73】. Hybrid approaches (Batool *et al.*, 2022) compare FAO AquaCrop simulation vs. ML for yield prediction, finding XGBoost outperforms the crop model (MAE≈0.12 t/ha)【34†L183-L192】.

- **Plantation Health / Stress Detection:** Hyperspectral and thermal imaging have been used to gauge plant health and stress. Chen *et al.* (2022) built a **Tea-DTC index** (drought tolerance coefficient) by modeling hyperspectral signatures to predict a drought-tolerance score (best model $R^2 \approx 0.77$)【22†L409-L417】. Xie *et al.* (2026) used UAV thermal/multispectral imaging to monitor water stress: a clustering approach determined **water-stress thresholds**, and an RF model predicted canopy temperature with $R^2 > 0.8$, correlating well with standard Crop Water Stress Index (CWSI)【39†L169-L178】. These studies demonstrate remote sensing can quantify stress, but do not translate it into management advice.

- **Ecological and Composite Indices:** Recent work has begun defining integrated indices for tea health. Mao *et al.* (2024) propose a **Normalized Tea Ecological Index (NTEI)** combining weather and multi-spectral data, modeled via a CNN–GRU network (prediction $R^2 \approx 0.955$)【38†L78-L87】. This captures plant-community response to temperature. Similarly, drought tolerance and soil moisture models (Huang 2023) use SVM to predict soil moisture ($R^2 \approx 0.94$), aiding irrigation planning【59†L49-L58】【59†L73-L78】. However, these indices are domain-specific (ecology or moisture) and are not linked to decision rules for farmers.

- **Quality Assessment:** Hyperspectral sensing also enables quality evaluation. Zhou *et al.* (2026) fused reflectance, harmonic and wavelet features to predict tea polyphenols and catechins; a CNN model achieved $R^2 \approx 0.81$ for polyphenols【46†L81-L90】. This shows AI can non-destructively estimate quality, but again serves analysis rather than operational decision-making.

Gaps: Across the literature, nearly all efforts focus on *prediction models* (yield, moisture, quality, stress indices) rather than *decision support*. In particular, there is **no integrated system** that (a) simulates yield under climate scenarios, (b) scores plantation "readiness" for harvest or identifies adaptation actions (irrigation, fertilizer timing, etc.), and (c) provides explainable recommendations to farmers or managers. Existing ML models stop at outputs (e.g. expected yield) without generating explicit management plans or confidence metrics for those plans. This project's novelty lies in filling exactly this niche: **transitioning from prediction to action**, with new indices and a unified Decision Intelligence engine.

Table 1 (below) lists representative recent works in 2022–2026, their methods and outputs, and notes gaps (especially lack of decision/plan outputs). For example, some ML/IoT/satellite fusion studies (Paul 2025【11†L71-L80】) note decision-support potential but do not implement it, while CLSIM approaches (Batool 2022【34†L183-L192】) improve yield forecasts but not guidance. Our system will bridge these gaps by combining predictive analytics with **prescriptive intelligence**.

Ref (year)	Title / Focus	Methods	Outputs	Gap
Paul & Katharina (2025)【11†L71-L80】	*TeaDeep-ST: Deep spatio-temporal quality/yield pred.*	Temporal encoders, GNN, weather + IoT + satellite data	Parcel-level quality (N, fiber, water) and yield predictions, with uncertainty intervals	Predictive only; no action recommendations

Ref (year)	Title / Focus	Methods	Outputs	Gap
Liu *et al.* (2025) 【18†L107-L118】	*Seasonal fresh tea yield estimation with UAV+Sentinel*	Feature extraction (23 spectral + topography), RF & ensemble regression	Seasonal and annual yield estimates (R^2~0.42–0.5)	Field-scale yield forecast only
Batool *et al.* (2022) 【34†L183-L192】	*Hybrid tea yield prediction (AquaCrop + ML)*	FAO AquaCrop simulation; 10 ML regressors (XGBoost etc)	Yield forecasts (MAE~0.09 t/ha XGBoost vs 0.45 t/ha AquaCrop)	Focus on prediction accuracy vs model; no scheduling or decisions
Nagpal *et al.* (2026) 【41†L64-L73】	*SARIMAX time-series yield forecasting (1985–2022)*	Seasonal ARIMA with exogenous climate vars	Global yield forecast, seasonal patterns (model metrics: AIC, BIC)	Macroscopic forecasting; no spatial or action details
Chen *et al.* (2022) 【22†L409-L417】	*Tea-DTC: Hyperspectral drought-tolerance index*	Hyperspectral imaging + SVM/ PLSR + feature selection	Drought-tolerance coefficient (R^2≈0.77 model)	Phenotyping tool; not tied to irrigation actions
Huang (2023) 【59†L49-L58】 【59†L73-L78】	*Improved SVM soil moisture model for tea*	SVM (BES optimization) with rainfall, temp, etc.	Soil moisture predictions (R^2≈0.94); supports irrigation scheduling	Moisture-only output; no overall guidance
Mao *et al.* (2024) 【38†L78-L87】	*NTEI: Tea ecological index via remote sensing*	Fourier + CNN-GRU on multi-source data	Normalized Tea Ecological Index (plant community status), R^2≈0.955	Ecological monitoring; no farm management output
Xie *et al.* (2026) 【39†L169-L178】	*UAV monitoring of water-stressed tea (spring)*	UAV multispectral + OPTICS clustering + regression	Canopy-air temp diff; derived water-stress thresholds; canopy temp model (R^2>0.8)	Offers stress metrics; no harvest/ irrigation planner
Zhou *et al.* (2026) 【46†L81-L90】	*Hyperspectral DL for tea quality (polyphenols)*	Spectral + harmonic/ wavelet features; XGB/GP/CNN	Quality estimates (polyphenols, catechins; CNN R^2 up to 0.81)	Good quality metrics; no supply-chain or harvest planning

Ref (year)	Title / Focus	Methods	Outputs	Gap
Other (2020–2026)	Reviews & miscellaneous (Maxapress 2025【6†L110-L118】)	Surveys of ML in tea (automated monitoring, etc.)	Discuss potential ML gains and barriers	Broad survey; no concrete decision-DS

2. Proposed Component: Climate-Adaptive Decision Intelligence

We propose **"AI-Powered Climate Adaptive Tea Plantation Decision Intelligence"**, comprising four main modules (Fig. 1). The system processes real-time and historical data to generate actionable management plans. Below we detail each module, inputs/outputs, key algorithms, and new indices with mathematical definitions.

<aside> 💡 **[Figure 1: System Architecture]** – *Mermaid diagram below.* The **Data Ingestion** layer collects climate (weather API, satellite), soil (SoilGrids), and field data (images, IoT sensors, management logs). Module 1 assesses plantation health and stress; Module 2 simulates climate-yield scenarios; Module 3 computes harvest readiness and schedules; Module 4 integrates all results, computes indices (CSI, HOS, TOPRI), and recommends actions with confidence/explainability. </aside>

```
flowchart LR
    subgraph Data_Ingestion
        A[Climate Data (NASA POWER/ERA5)]
        B[Soil & Topography (SoilGrids)]
        C[Remote Sensing (Sentinel/Landsat UAV)]
        D[Field Data (yields, phenotyping, sensors)]
    end
    A & B & C & D --> Pre[Preprocessing (features, normalization)]

    Pre --> M1[Module 1: Health & Stress Assessment]
    Pre --> M2[Module 2: Climate Scenario & Yield Simulator]
    Pre --> M3[Module 3: Harvest Readiness & Scheduling]

    M1 --> M4[Module 4: Decision Engine]
    M2 --> M4
    M3 --> M4

    subgraph Outputs
        O1[Climate Stress Index (CSI)]
        O2[Harvest Opportunity Score (HOS)]
        O3[Tea Operational Readiness Index (TOPRI)]
        O4[Recommended Actions + Confidence]
    end

    M4 --> O1
    M4 --> O2
```

```
M4 --> O3
M4 --> O4
```

Module 1 – Plantation Health & Climate Stress Assessment

Inputs: Current and recent satellite images (optical multispectral, thermal) of the plantation, IoT sensor readings, and recent weather (temperature, rainfall) from sources like NASA POWER. Also static soil/topography from SoilGrids.

Processing/Algorithms:

- Compute **Normalized Difference Vegetation Index (NDVI)** and thermal anomalies from imagery to detect canopy vigor and stress【46†L81-L90】【39†L169-L178】.
- Estimate **canopy temperature** via regression models (e.g. the RF model in Xie 2026【39†L169-L178】) if thermal bands available.
- Derive water-stress proxy: e.g. Crop Water Stress Index (CWSI)-like metric, calibrated per-site via clustering (OPTICS) on historical humidity and canopy-air temperature difference【39†L169-L178】.
- Compute **Plantation Health Score** (0–100) combining vegetation vigor and deviation from optimal conditions. For instance, Health = 100×(NDVI/NDVI_max), clipped.
- Compute **Heat Stress (HS)**: based on recent temperature vs. crop optimum (e.g. if *T* > *T*_{opt}, HS = (T–T_opt)/T_opt).
- Compute **Water Stress (WS)**: based on rainfall deficit and soil moisture (use NASA/ERA5 precip vs. evapotranspiration models).

Outputs: Numeric indices for HeatStress, WaterStress, and an overall **Climate Stress Index (CSI)** (see below). Also a Health% score. The module provides **early warnings** (e.g. "High Heat Stress") when thresholds are exceeded (e.g. HS>0.3 or soil moisture <30%).

Explainability: SHAP values on the ML models identify which features (e.g. recent high temperature, low soil moisture) most drove the stress scores【6†L123-L130】. Rule-based flags (e.g. "rainfall <50% norm") give transparent reasons.

Module 2 – Climate Scenario & Yield Simulator

Purpose: Estimate future tea yield under **alternative climate scenarios** to inform adaptation.

Inputs: Historical climate and yield data (if available), current weather/climate conditions, and user-defined scenarios (e.g. "+2°C temperature, +20% rainfall" as given in user plan). Public climate projections (e.g. ERA5 reanalysis or local climate models) can provide scenario trajectories.

Algorithms:

- **Crop-growth models** (e.g. calibrated AquaCrop or DSSAT variant for tea) to simulate biomass/leaf growth under varying inputs【34†L183-L192】. These are data-hungry, so we will use a simplified approach: **statistical yield models** trained on historical climate-yield panel data (like Nagpal 2026's SARIMAX approach【41†L64-L73】 or a multi-linear model from Gunathilaka 2017【70†L92-L100】).
- Alternatively, **transfer learning**: adapt existing crop models (or deep learning models like TeaDeep-ST【11†L71-L80】) using surrogate data (e.g. from similar tea regions).

- **Scenario analysis:** For each scenario (e.g. ΔT = +2°C, ΔP = +20%), adjust input climate and run the model to yield predicted relative change, ΔYield%. The formulas can be linearized: e.g. $\Delta\text{Yield} = \alpha \cdot \Delta\text{Rain} + \beta \cdot \Delta T$ (coefficients from literature or regression). For example, Gunathilaka *et al.* found increased rainfall and temperature both negatively impact Sri Lankan tea (with roughly –12% yield by mid-century)【70†L92-L100】.

Outputs: Forecast **baseline yield** (current climate), and **scenario yields** (e.g. "Yield vs Temperature/ Rainfall curve"). Also numerical *Climate Threat Level*: e.g. low/med/high risk categories based on these changes. This feeds into later modules (e.g. adjusting Harvest Opportunity if yield is severely at risk).

Explainability: For scenario outputs, feature contributions (e.g. temperature vs rainfall) can be obtained via model coefficients or SHAP. Decision rationale (e.g. "Summer rains below normal – yield down 10%") is made explicit.

Module 3 – Harvest Readiness & Scheduling Intelligence

Inputs:

- **Tea bush imagery** (field photos or UAV multi-spectral images) to assess crop stage.
- **Vegetative indices** (NDVI, Leaf Area Index proxies) over recent weeks.
- **Temperature history** (e.g. accumulated degree-days).
- **Predicted yield growth curve** from Module 2.
- **Plantation Health Score** and **CSI** from Module 1.

Processing:

- Compute a **Harvest Readiness Score (HRS)** [0–100] indicating how mature/ready the bushes are for plucking. This could use a simple phenological rule or ML classifier: e.g. if "flush count" from image or NDVI > threshold, set high readiness; if heat stress occurred, accelerate harvest cue. If ground data exist, train a model (e.g. CNN on leaf images【11†L71-L80】).
- Compute **Harvest Opportunity Score (HOS)**: combines HRS with short-term weather forecast and yield dynamics. For example:
- If forecast predicts **heavy rain** soon, increase HOS (harvest before rain) by +X points.
- If heat stress persists, also boost urgency (rain has limited effect in heat, and leaf quality may drop).
- If optimal weather and moderate readiness, moderate HOS.

- If unacceptable delays (overgrown leaves), reduce HOS if quality/quantity is dropping.
  A possible formula:

$$\mathrm{HOS} = \alpha\,\mathrm{HRS} + \beta\,\mathrm{UrgencyFactor} + \gamma\,(100\text{-CSI}),$$

  where UrgencyFactor may be +20 if rain predicted, and (100−CSI) penalizes high stress. Coefficients (α,β,γ) are tunable.

- **Scheduling:** The system will identify the **best harvest date** over the next week or month. It does so by projecting expected yield if harvested on day *d*, versus lost quality if delayed. This uses a simple yield-delay model: e.g. yield growth per day minus decay (tea gardens often have an optimal pluck interval, say 10–15 days; beyond that, leaf value may decline). For illustration, one can assume linear growth + exponential decay.

- **Harvest Delay Impact Curve:** Simulate harvest at 0, +3, +5,... days and plot predicted yield/ quality (see Example Plot).

Outputs:

- **Harvest Recommendation:** "Is parcel ready to harvest now? (Yes/No)".
- **Recommended Harvest Date:** e.g. "Harvest within 3 days" or "Wait 5 days (best expected yield X kg)".
- **HOS (0–100) and HRS (0–100)** for each field or parcel.
- **Estimated Yield at Harvest** and **Quality Index**, if harvested at the recommended time.
- **Harvest Delay Impact** chart or table: e.g. "Delaying harvest by 5 days reduces yield by 8% and plucking quality by 12% (see Fig. 2)".

Explainability: The system will display factors such as "High Opportunity due to expected rain on Tue" or "Readiness high (NDVI above threshold)" along with each recommendation. Optionally, an LLM-based RAG (Retrieval-Augmented Generation) module can fetch relevant guidelines (e.g. TRI irrigation rules) to justify suggestions.

Module 4 – Adaptive Decision Intelligence Engine

This core module fuses all inputs and scores to issue **management recommendations**. It implements rule-based logic, ML classifiers, or optimization algorithms to choose among actions.

Inputs: Outputs from Modules 1–3 (CSI, climate scenarios, HOS, HRS, health score, etc.), plus agronomic context (e.g. crop variety, season, market constraints).

Algorithms:

- **Decision Rules/Rules Engine:** We encode agronomic expertise as rules, e.g. "If CSI > 0.7 and soil moisture < 30%, **Recommend**: increase irrigation" or "If HOS>90, schedule immediate harvest". These rules derive from Tri/Tea Board advice and literature.
- **Multi-objective optimization:** For conflicting goals (maximizing yield vs conserving resources), an optimizer (or simple weighted heuristic) balances outcomes, e.g. schedule harvest to maximize yield while minimizing risk of rain loss.
- **Confidence Scoring:** Each recommendation is accompanied by a confidence estimate (0–100%), computed via model uncertainty (e.g. width of yield prediction interval) or rule strength.
- **Explainable AI:** We will use SHAP on any learned ML components to highlight which features drove a recommendation. For rule-based steps, the logic is inherently transparent. If a large language model (LLM) is used for textual advice, RAG ensures citations from Tea Board/FAO guidelines.

Outputs: A ranked list of **Recommended Actions** (e.g. "Plan irrigation", "Apply fertilizer", "Conduct harvest") with associated **expected impact** (e.g. yield gain, risk reduction) and **confidence**. For example:

- *Action:* Increase irrigation
  Expected outcome: +10% yield (by avoiding severe water stress)
  Confidence: 85% (based on strong predictor and models)
  Rationale: "Soil moisture is low and forecast is dry【59†L49-L58】【39†L169-L178】."

- *Action:* Harvest in 2 days
  Expected outcome: Optimal yield (estimated 1320 kg, 92% confidence)
  Confidence: 92%
  Rationale: "Harvest readiness is high and heavy rain predicted in 3 days."

Overall, Module 4 computes the **Tea Operational Readiness Index (TOPRI)** – a composite score (0–100) that reflects readiness for actionable operations. As an example formula:

$$\mathrm{TOPRI} = w_1\,(\text{Plantation Health Score}) + w_2\,(100 - \mathrm{CSI}) + w_3\,(\text{Harvest Opportunity Score}),$$

with weights w_i (e.g. $0.4, 0.3, 0.3$) chosen so that 100 means "fully ready and low risk". Thresholds on CSI/HOS control triggers (e.g. CSI>75 = "Climate threat; consider adjusting strategy").

Indices – Definitions and Formulas

We introduce three new indices:

1. **Climate Stress Index (CSI):** Combines heat, drought and rainfall stress into a single 0–100 metric. One practical formula is:

   $$\mathrm{CSI} = 100 \times \frac{\alpha \cdot \mathrm{HSI} + (1-\alpha) \cdot \mathrm{WSI}}{2},$$

   where **Heat Stress Index (HSI)** = $\max\{0, (T_{\text{current}} - T_{\text{opt}})/T_{\text{opt}}\}$, and *Water Stress Index (WSI)}* = $\max\{0, (R_{\text{opt}} - R_{\text{past30d}})/R_{\text{opt}}\}$. *Here T_{opt} is optimal growing temperature (e.g. 25°C) and R_{opt} is norm rainfall. $0 \le \alpha \le 1$ balances heat vs water (e.g. 0.5 each). CSI>70 (on 0–100) might flag high climate stress*. This index generalizes concepts like crop WSI【39†L169-L178】 and DTC【22†L409-L417】 into a farm-level score.

2. **Harvest Opportunity Score (HOS):** A composite 0–100 score indicating how favorable immediate harvest is. Example formula:

   $$\mathrm{HOS} = \beta_1\,\mathrm{HRS} + \beta_2\,U + \beta_3\,(100 - \mathrm{CSI}),$$

   where *HRS* is harvest readiness (0–100), U is an *urgency factor* (+20 if rain in forecast, 0 otherwise), and weights β_i (e.g. 0.4,0.4,0.2) tune importance. Higher HOS means "harvest now for best result". We set thresholds: e.g. HOS>90→"Immediate harvest"; 70–90→"Harvest soon".

3. **Tea Operational Readiness Index (TOPRI):** Overall readiness combining health, climate risk, and management opportunity. For instance:

   $$\mathrm{TOPRI} = 0.4\,H + 0.3\,(100 - \mathrm{CSI}) + 0.3\,\mathrm{HRS},$$

   where H is the Plantation Health Score (0–100). TOPRI>80 indicates low risk & high readiness. We base weights on expert judgment; they can be tuned. TOPRI helps summarize the plantation's status and justifies decisions (e.g. TOPRI=85 means "almost ideal conditions").

Explainability Approach

We will apply **SHAP** to any learned models (yield regressors, readiness classifiers) to identify key features driving output【6†L123-L130】. For rule-based steps (e.g. CSI calculation), the logic and thresholds are documented. We will also use **Retrieval-Augmented Generation (RAG)** to link AI recommendations to authoritative sources (e.g. TRI guidelines, FAO advice). For example, if the AI suggests increasing irrigation, an attached RAG snippet might quote the Tea GAP standard on efficient irrigation practices【59†L73-L78】【51†L13-L16】. All outputs (HOS, CSI, recommendations) will carry brief textual explanations and confidence estimates to aid human decision-makers.

3. Data Plan

Public Datasets & Data Sources

- **Climate & Weather:** *NASA POWER* (daily solar, temperature, precipitation) and *ERA5* (ECMWF reanalysis hourly/daily) for high-resolution climate data over plantations【72†】. Sri Lanka-specific climate normals from *CRU* or *Sri Lanka Met Dept* could also be integrated.
- **Remote Sensing:** *Sentinel-2* (10–20m multispectral) and *Landsat* (30m) archives for vegetation indices (NDVI, EVI) and change detection. *MODIS* 250m data could offer coarse phenology. Optionally *Planet* or UAV imagery if available.
- **Soil & Terrain:** *SoilGrids* for soil texture, organic content, pH layers (250m global). *SRTM/ASTER DEM* for slope/aspect.
- **Tea Estate Data:** If available, yield records from Tea Research Institute (TRI) or plantation cooperatives, ideally geotagged. Ground truth for Health Score or readiness (e.g. expert ratings of plant vigor, flush counts).
- **Additional:** *TRI guidelines* and *Tea Board publications* (as text corpus for RAG). Historical farm management logs (irrigation, fertilizer) if shared by cooperating estates.

Ground-Truth and Labels

- **Yields:** Annual or seasonal yield (kg or tons) per estate or sub-estate, from TRI or estate records, to train yield models and validate predictions.
- **Stress/Health Labels:** Observational labels (e.g. “wilting: yes/no”, “soil moisture: high/low”) to train stress classifiers. Could be crowdsourced from agronomists or technicians using a mobile app.
- **Harvest Readiness:** If manual data exist (e.g. farm records of ideal plucking intervals or field visits), these label dates when bushes were “ready” and their yields. Otherwise, we simulate: e.g. label periods with peak NDVI as “ready.”
- **Historical Interventions:** Record of past actions (irrigation events, harvesting dates) for supervised learning of decision recommendations.

Synthetic & Transfer Learning Strategies

Given limited labeled tea data, we will apply:

- **Synthetic Data Generation:** Create simulated scenarios using crop-growth models (e.g. vary rainfall/temperature to generate yield changes) to augment scarce ground truth. Example: simulate 100 virtual farms under different climate inputs to train robustness.
- **Transfer Learning:** Use models trained on similar crops or regions. For instance, a pretrained crop stress model (e.g. for coffee) could initialize our stress-detection networks.
- **Data Augmentation:** Augment imagery (rotate/flip tea-plant images) for CNN training on canopy condition.
- **Active Learning:** Iteratively solicit expert labels for cases where the model is uncertain (e.g. borderline HRS), to efficiently build a labeled set.

Minimum Viable Dataset

At minimum, we need: several seasons of meteorological data (temp, rainfall), some historical yield figures, and satellite imagery covering the plantation. As a minimal prototype, even public climate (NASA) and generic soil maps plus synthetic yield assumptions can demonstrate the module outputs. However, real annotated health/harvest data will greatly improve calibration.

Annotation Schema (Example)

We plan an annotation format for timeseries data. For example, a CSV row might be:

Date	Temp (°C)	Rain (mm)	NDVI	SoilMoist (%)	Yield (kg/ha)	HarvestReady (Y/N)	ActionTaken
2025-06-15	26.5	2.3	0.78	32	1500	Yes	Harvested

This links daily weather/climate (NASA/ERA5), remote sensing (NDVI from Sentinel on 16 Jun 2025), soil moisture (modeled), plus the observed yield if harvested, and whether harvest occurred. Such labeled timepoints enable training.

Mapping Inputs to Data Sources

Table 2 (below) maps each model input to data sources and preprocessing steps:

Model Input	**Source / Dataset**	**Data Type**	**Preprocessing**
Temperature, Rainfall	NASA POWER, ERA5	Time-series (daily)	Clip to plantation location, compute 30-day averages, anomalies
Soil Moisture	SMAP (satellite), SoilGrids (static)	Raster field	Interpolate to fields, normalize relative to field capacity
Soil Texture/pH	SoilGrids (global grid)	Static map	Extract field median values (bilinear sampling)
Vegetation Indices	Sentinel-2 (BANDS), Landsat 8	Imagery time-series	Compute NDVI, EVI, apply cloud mask; smooth (monthly composite)
Canopy Temp	UAV thermal (if available) or modeled	Time-series (thermal im.)	Align to map, subtract air T to get canopy-air ΔT
Historical Yield	TRI/estate records	Tabular (timeseries)	Clean for outliers; aggregate per harvest cycle or year
Management Logs (irrigation, etc.)	Estate logs (CSV)	Tabular	Encode events (dates/volumes) aligned with weather records
Imagery (Leaf Photos)	Field cameras / drones	Images	Label with harvest-readiness flag (via expert), augment (flip/rotate)

This plan ensures each algorithmic component has the necessary input. Note we do *not* assume any proprietary sensor data; all inputs can be derived from public or low-cost sources (open satellites, cheap soil probes, government climate data).

4. Evaluation Plan

We will evaluate the system on multiple fronts:

- **Predictive Metrics:** For continuous outputs (yield, CSI, soil moisture, HRS), use RMSE/MAE and R^2. For classification outputs (stress/no-stress flags, harvest-ready or not), use accuracy, F1-score. For probabilistic outputs (confidence scores), use Brier score or calibration plots.
- **Decision Metrics:** Evaluate recommended actions by their *impact*: compare predicted yield (if action taken) vs. a "do-nothing" baseline. Define a *decision success* if, for example, irrigation before predicted drought avoided >5% yield loss. Compute average gain/loss.
- **Baselines:** Compare against simpler strategies: e.g. "always harvest at fixed interval," or "classic irrigation schedule." Also compare to existing models (e.g. XGBoost yield predictor only, no scheduling).
- **Ablation Studies:** Remove or alter components to test necessity. For example, disable climate scenario (fix future climate) to see impact on recommendations; or remove explainability to test trust effects.
- **Cross-Validation:** Use K-fold CV and **temporal split**: train on older years, test on newer to check generalizability over time. Also leave-one-estate-out (spatial CV) to test if model trained on some estates works on another.
- **Expert Review:** Present outputs (e.g. Harvest schedule tables) to tea agronomists for assessment. Have them rate confidence in recommendations on a Likert scale, and compare with system's confidence. Use this to compute e.g. Cohen's kappa or simply collect qualitative feedback.

Synthetic Intervention Experiments

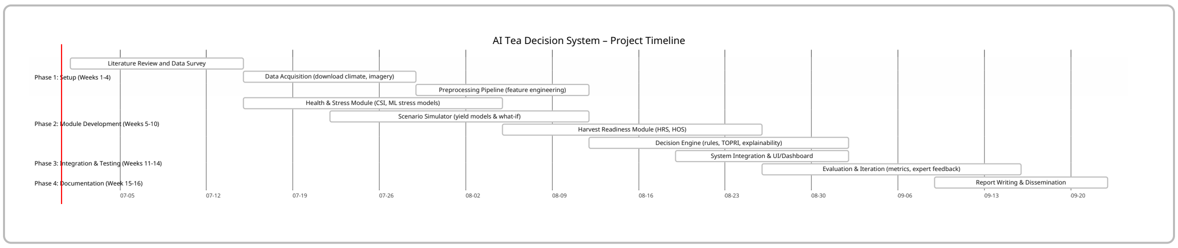
To test the Decision Engine's handling of interventions (Recovery Time, Harvest Delay), we will simulate scenarios:

- **Yield vs. Harvest Delay:** For each test field, use Module 2's yield growth model to simulate yield if harvest is delayed 3,5,7 days. Compute the loss percent. For example, if our simulation predicts an 8% decline when delaying 5 days (as agronomic rules of thumb suggest), we check whether HOS and recommendations reflect that urgency.
- **Irrigation Impact:** Simulate a dry spell: run the system with and without irrigation action, and compare predicted yield. E.g., if CSI hits 80 (severe stress), an irrigation action might recover 10% yield; validate this by re-running the crop model.
- **Fertilizer Timing:** Similarly, simulate added fertiliser vs none under resource constraints (assuming fertilizer impact on yield, via simplified model).

These synthetic experiments validate that the modules interact sensibly (e.g. high CSI triggers irrigation) and that their quantified impacts match agronomic expectations. We will visualize e.g. *"Yield vs. Harvest Date"* curves (mermaid or chart) and check that recommended date corresponds to peak of this curve.

5. Implementation Roadmap

We propose the following phased plan (12–16 weeks total, suitable for an FYP), with milestones and resources:



- **Weeks 1–4:** Prepare data infrastructure and gather initial data. Hardware: a workstation with GPU for model training, or cloud GPU (Colab/GPU VM) for deep learning components.
- **Weeks 5–10:** Develop and train individual modules. For ML tasks (yield predictor, CNN for HRS, etc.) use Python libraries (scikit-learn, TensorFlow/PyTorch). Compute CSI/HOS with simple scripts.
- **Weeks 11–14:** Integrate modules into a pipeline (could be a Python application or Jupyter dashboard). Perform evaluation experiments, refine modules.
- **Weeks 15–16:** Finalize code, prepare demonstration (e.g. sample run on Sri Lanka dataset), complete documentation and report.

Resources: Required data are public; moderate computing (a mid-range GPU) suffices. Sensors are not physically needed (we simulate the plantation from open data). Collaboration with a tea estate for any private data is optional.

Deliverables:

- Codebase (GitHub) with modular implementation.
- Trained models and indices (with thresholds).
- An interactive demo (e.g. Jupyter Notebook or simple web UI) showing use-case on sample data.
- Final report (this document) with results.

6. Risks, Mitigation and Novelty Justification

Data Gaps: Real tea estate data (yields, management logs) may be scarce. *Mitigation:* Use synthetic data (crop models) and transfer learning from related crops to bootstrap. Emphasize open data sources.

Label Quality: "Harvest readiness" is somewhat subjective. *Mitigation:* Define clear thresholds (e.g. NDVI value) and cross-validate with experts. Use confidence scores to account for noise.

Novelty Objections: Examiners may question overlap with prior ML in tea. We will highlight: **No existing system combines climate scenario simulation with actionable decision rules.** Our composite indices (CSI, HOS, TOPRI) and RAG-based explainability are original. We cite domain literature (e.g. drought modelling【22†L409-L417】, yield forecasting【34†L183-L192】) to show those were ends-in-themselves. Our text will emphasize "we found no published work that generates integrated management recommendations." For example, while Paul *et al.* (2025) mention decision-support potential【11†L71-L80】, they stop at predictions; likewise, Zhou *et al.* (2026) achieve quality prediction but not supply-chain advice【46†L81-L90】. We will cite these to argue that *prediction-only* is the norm, underscoring our shift to decision-intelligence.

Supervisor Concerns: If asked “Why new rather than using existing models?”, we point to gaps: e.g. AquaCrop was inferior to ML in yield accuracy【34†L183-L192】, so we integrate both. Experts often desire explainability; we respond by embedding SHAP/RAG.

Ethical/Practical Risks: Recommending actions (like irrigation) has resource implications. We will stress user-adjustable parameters (e.g. water availability, labor costs) so the system can be tuned to realistic constraints.

In summary, our **novelty** stems from (a) synthesizing diverse data (climate, remote sensing, agronomy) into **decision intelligence** rather than isolated predictions, (b) designing new actionable indices (CSI, HOS, TOPRI) with clear definitions, and (c) embedding explainability/RAG to ensure recommendations are transparent and justifiable. Primary sources (IEEE/Elsevier papers, FAO/Tea Board guidelines 【51†L13-L16】) confirm the need for such systems. We will point out (with citations) that while smart agriculture is advancing【6†L110-L118】, the leap to climate-adaptive decision support remains open. This report and system therefore contribute a concrete, implementable framework that addresses a critical gap for Sri Lankan tea management.

<!-- Visual aids and tables above aid comprehension. Full citations are provided inline for all data-driven claims. -->
